

Name:		Index Number:		Class:	
--------------	--	----------------------	--	---------------	--



DUNMAN HIGH SCHOOL **Preliminary Examination** **Year 6**

H2 BIOLOGY

9648/02

Paper 2 Structured and Free-Response Questions

15 September 2015

BOOKLET A

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section B Structured Questions

Answer **all** questions on **both** booklets **A** and **B**.

Write your answers on space provided in the Question Paper.

Section C Free-Response Questions

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate.

Answer each part (a) and (b) on a fresh piece of writing paper.

Submit your answers to Sections B and Section C separately.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

For Examiner's Use	
Section A [40]	
Section B [80]	
1	/ 10
2	/ 12
3	/ 7
4	/ 5
5	/ 6
6	/ 15
7	/ 15
8	/ 10
Section C [20]	
1 / 2	
Total [140]	

This document consists of **10** printed pages and **2** blank page.

[Turn over

Section B: Structured Questions (80 marks)
 Answer **all** questions in this section.
Booklet A

*For
Examiner's
use*

Question 1

Fig. 1.1 below is an electron micrograph showing part of an animal cell.

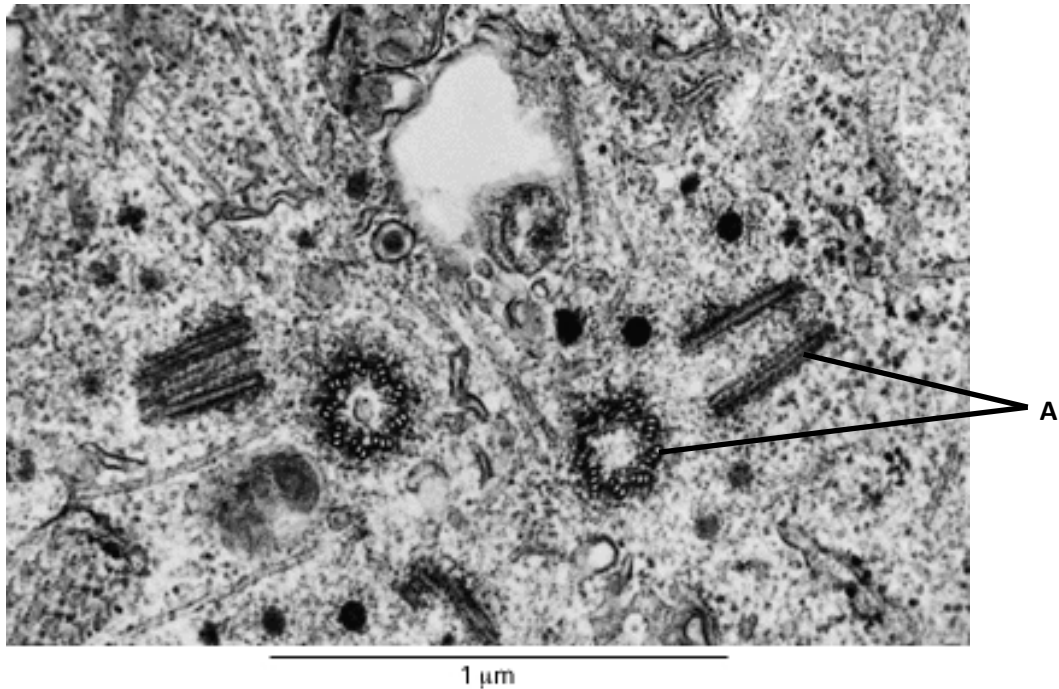


Fig1.1

- (a) (i) Name the structures labeled A. [1]

A :

- (ii) Describe **two** structural differences between structure **A** and cellulose. [2]

- (iii) Name the process the cell is undergoing and explain your answer. [2]

- (b) Changes in DNA amount and chromosome numbers were monitored during one cycle of meiosis. Results are shown in Fig. 1.2.

For
Examiner's
use

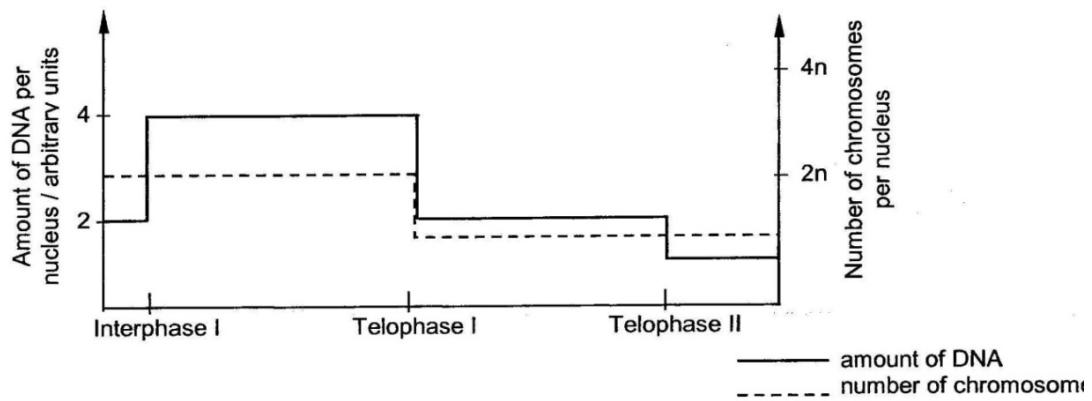


Fig1.2

- (i) Explain why DNA amount and chromosome numbers remain constant during the two anaphases in Fig. 1.2. [2]

- (ii) Explain the significance of meiosis in genetic variation. [3]

Total:[10]

Question 2

- (a) Hydrolase is a group of enzymes which catalyse the hydrolysis of chemical bonds and can be found in plants. Describe the roles of the proteins involved in hydrolase mRNA synthesis. [2]

Fig. 2.1 shows the process of hydrolase synthesis.

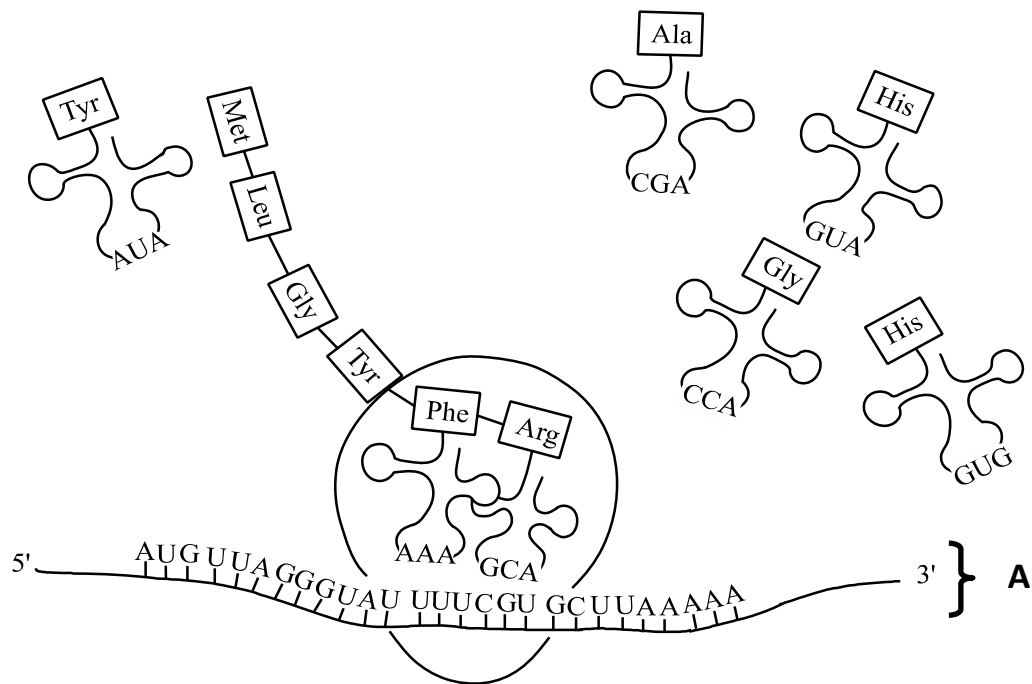


Fig. 2.1

- (b) Describe the role of **A** in the synthesis of hydrolase. [3]

- (c) Hydrolase can be found in lysosome. Action of hydrolase can lead to autolysis by causing conformational change to the shape of ribosome. Describe the action of hydrolase on ribosomes and how this could cause cell death. [3]

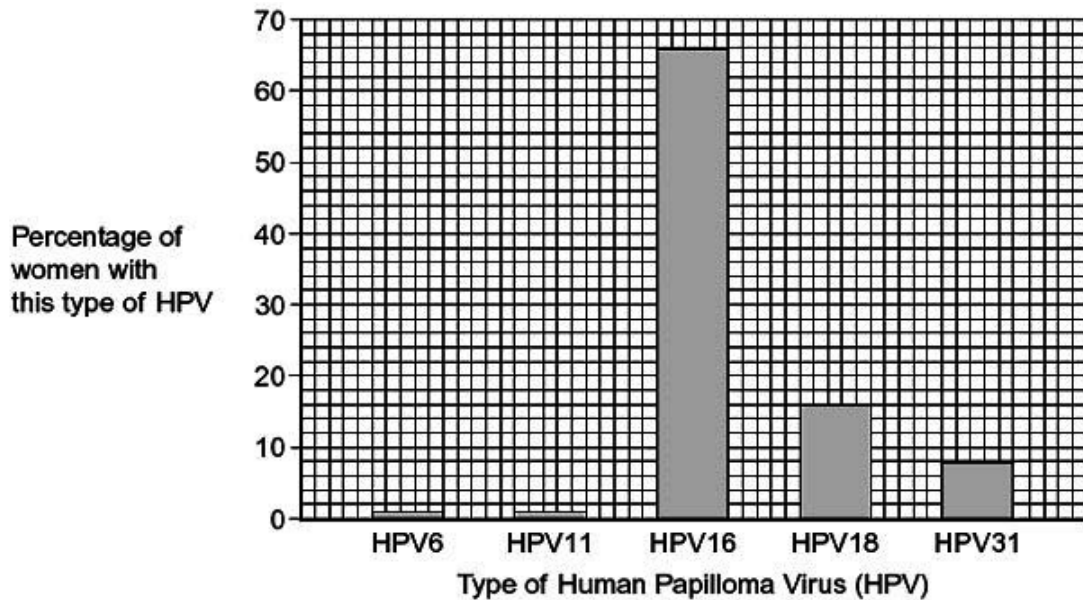
- (d) Plant glycoside hydrolases is involved in cell wall polysaccharide degradation. Suggest two stages during plant development when cell wall composition changes. [2]

- (e) Hydrolase can also be found in human. The human genome contains approximately 300 hydrolase genes which are able to encode a reported 1200 different types of polypeptides reported to exist in the human cell. Explain how this is possible. [2]

Total:[12]

Question 3

Cervical cancer occurs in the cervix. The cervix is the opening of the uterus. Scientists investigated the link between cervical cancer and infection with some types of Human Papilloma Virus (HPV). **Fig. 3.1** shows the frequency of five different types of HPV in women who had cervical cancer.

**Fig. 3.1**

- (a) A local newspaper published an article about cervical cancer with the headline “HPV causes cervical cancer.” Using data from **Fig 3.1**, comment if this claim may be supported. [1]

A study published in the 28th edition of *AIDS* has shown that HIV-positive women are more likely to carry multiple strains of human papilloma virus (HPV). It has been proven that early intervention by antiviral drugs can prevent HIV.

- (b) Suggest an explanation for the high incidence of HPV in HIV patients. [2]

- (c) Suggest and explain the possible modes of action of the antiviral drugs in halting the reproductive cycle of HIV. [4]

*For
Examiner's
use*

Total: [7]

Question 4

For
Examiner's
use

Fig. 4.1 shows the structure of a telomere.

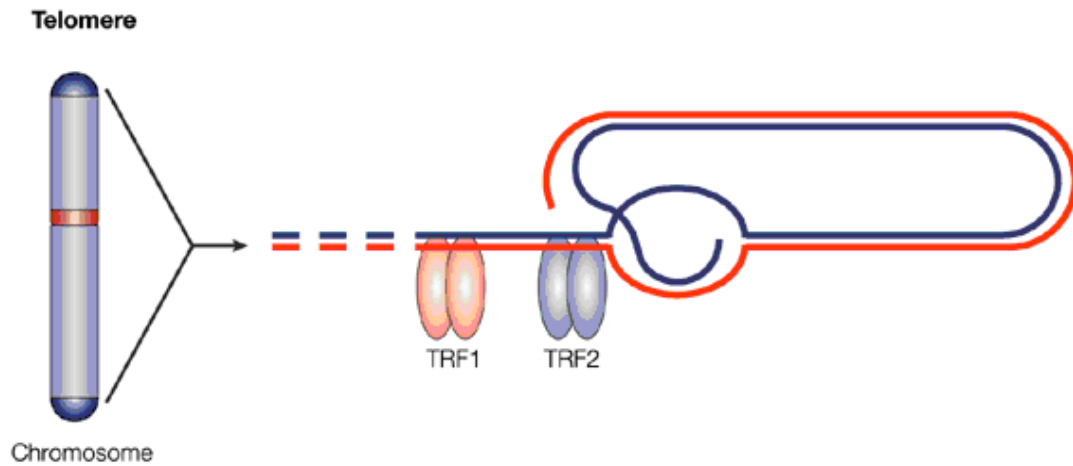


Fig. 4.1

(a) (i) Label on the diagram, the 3' and 5' ends of the DNA. [1]

(ii) Suggest why the structure of the telomeric end consists of a unique 'T loop'. [2]

(b) Suggest why bacteria do not have telomeres. [2]

Total: [5]

BLANK PAGE

Question 5

In the pea plant (*Pisum sativum*), the timing of flowering is controlled by multiple alleles. Investigators have found evidence for four different alleles at this flowering locus that can exert temporal control of flowering:

- F^V very early flowering
- F^E early flowering
- F^M mid flowering
- F^L late flowering

F^L is dominant over F^M , F^E and F^V .

F^M is dominant over F^E and F^V .

F^E is dominant over F^V .

Another **A/a** gene locus on a different chromosome codes for the colour of the flower which is either purple or white.

Table 5.1 below shows the results of crosses between a mid flowering plant with purple flowers with three other different plants:

Cross	Parents	Offspring
1	mid flowering, purple flowers x early flowering, purple flowers	50% mid flowering, 25% early flowering, 25% very early flowering; All purple flowers
2	mid flowering, purple flowers x mid flowering, white flowers	75% mid flowering, 25% early flowering; 50% purple flowers, 50% white flowers
3	mid flowering, purple flowers x late flowering, purple flowers	50% late flowering, 25% mid flowering, 25% early flowering; 75% purple flowers, 25% white flowers

- (a)** Draw a genetic diagram to explain the results of Cross 3. [4]

*For
Examiner's
use*

- (b)** Describe two differences between inheritance caused by multiple alleles and polygenic inheritance for a characteristic. [2]

Total: [6]

PROCEED TO BOOKLET B

BLANK PAGE



**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2014
YEAR SIX
H2 BIOLOGY (9648)
PAPER 2 (Booklet A)**

Structured Questions Answers

1(a)

(i) centrioles

(ii)

made of amino acids/tubulin vs β -glucose;

many different types of amino acids vs one type of β -glucose;

microtubules arranged in nine triplets vs many chains crossed-linked/joined together (to form microfibrils);

occur in pairs vs chains grouped to form macrofibrils; peptide bonds vs glycosidic bonds;

hollow cylinder vs solid/compact/non-hollow structure;

(iii)

Cell division /mitosis /meiosis / prophase (I/II)/ interphase;

R! metaphase, anaphase

There are two pairs/sets of centrioles in the cell;

R! '2' alone

(b)(i)

during anaphase I & II, even though homologous chr & sis chromatids are moving towards pole respectively;

the genetic material still remain within the same cell, nucleus is not formed yet, cell has not divided into 2 indiv cells;

so content & chr no. will remain unchanged/will remain const

(ii)

prophase I, crossing over b/w non-sis chromatids of homologous chr forming chiasmata, exchange of segments of chr result in new combination of alleles in gametes formed;

metaphase I & anaphase I, independent assortment and random segregation of homologous chr gives rise to diff combinations of chromosomes in gametes;

metaphase II & anaphase II, independent assortment and random segregation of non identical sis chromatids results in gametes having diff combination of alleles;

2(a)

RNA polymerase – Adds RNA nucleotides in a 5' to 3' direction of newly synthesized mRNA strand;

(General) Transcription factor – Binds to promoter to recruit RNA polymerase to form transcription initiation complex to initiate transcription;

Specific transcription factors (activators / repressors) – Regulates transcription by binding to enhancers or silencers;

2 max

(b)

A (mRNA) serves as a template for synthesis of hydrolase;

Information encoded as a triplet base / codon per amino acid;

Small ribosomal subunit attaches to the 5' end of A;

Complementary base pairing between the anticodons on tRNA and codons on mRNA during translation;

(c)

Hydrolase catalyses the hydrolysis of / break chemical bonds which keep ribosome 3D structure ideal for translation;

Thus ribosome conformation changes and will not be able to bind to mRNA / tRNA thus will not be able to translate;

Leads to loss of protein essential for cell survival e.g. respiratory enzymes;

(d)

cell division/expansion/ differentiation / in response to environmental stresses

(e)

- This is due to alternative splicing;
- Different pre-mRNA can be spliced into different mature mRNA by the inclusion of different sets of exons/ constitutive exons are always retained while different alternative exons are retained to form the mature mRNA;
- The resulting mature mRNAs which exit to the cytoplasm hence some differences in terms of the codons contained which translates to different polypeptide sequences;
- Accept: many genes have more than allele, and each allele codes for slight difference in polypeptide sequence.

3(a)

No, only 92% of all women with cancer has HPV of any type;
8% of women with cancer did not have HPV OR / control group / HPV in healthy women not studied / individuals may also possess more than 1 type of HPV;

Or

Yes, a large percentage of 66% of women have HPV 16 leading to cancer;
1 max

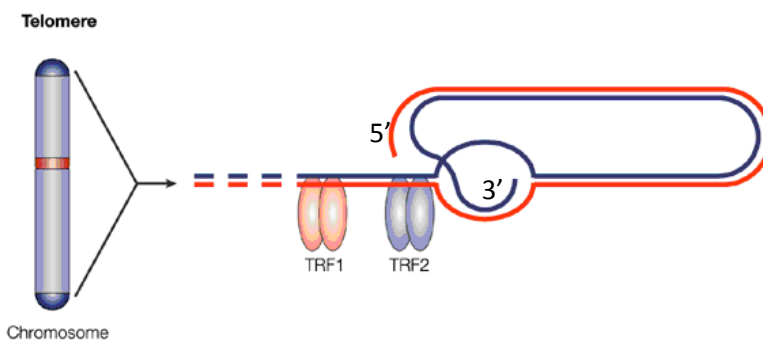
(b)

HIV infects T cells;
immune system compromised/ not effective;

(c)

drugs can bind to glycoproteins/ receptor molecules on CD4, prevent gp120 on HIV viral envelope from binding and entering cell;
drugs can bind to reverse transcriptase, prevent the synthesis of a single DNA strand that is complementary to the viral RNA;
drugs can bind to/ inhibit integrase, prevent the ds viral DNA from integrating as provirus into host cell chromosome;
drugs bind to/ inhibit HIV protease, resulting in viral proteins not being cleaved properly;

4 (a)(i)



(a)(ii)

single-stranded 3' overhang invade into / bind to the double-stranded telomeric DNA;

to prevent exonucleases from degrading the ends of the linear DNA molecules;

or

to prevent fusions of the ends of chromosome with other DNA molecules;

(b)

Bacteria have circular DNA;
ref to no gap left after removal of the primer at the 5' end of the leading strand, DNA polymerase fill up the gap;
No loss of chromosome ends at each division/ DNA replication;

5(a)

Parental phenotype: mid flowering, late flowering,
 purple flowers x purple flowers

Parental genotype: $F^M F^V Aa / F^M F^E Aa$ x $F^L F^E Aa$;

Gametes $(F^M A)$ $(F^M a)$ $(F^V A)$ $(F^V a)$ $(F^L A)$ $(F^L a)$ $(F^E A)$ $(F^E a)$;

gametes	$F^L A$	$F^L a$	$F^E A$	$F^E a$
$F^M A$	$F^L F^M AA$ late flowering, purple flowers	$F^L F^M Aa$ late flowering, purple flowers	$F^M F^E AA$ mid flowering, purple flowers	$F^M F^E Aa$ mid flowering, purple flowers
$F^M a$	$F^L F^M Aa$ late flowering, purple flowers	$F^L F^M aa$ late flowering, white flowers	$F^M F^E Aa$ mid flowering, purple flowers	$F^M F^E aa$ mid flowering, white flowers
$F^V A$	$F^L F^V AA$ late flowering, purple flowers	$F^L F^V Aa$ late flowering, purple flowers	$F^E F^V AA$ early flowering, purple flowers	$F^E F^V Aa$ early flowering, purple flowers
$F^V a$	$F^L F^V Aa$ late flowering, purple flowers	$F^L F^V aa$ late flowering, white flowers	$F^E F^V Aa$ early flowering, purple flowers	$F^E F^V aa$ early flowering, white flowers

;

Phenotypic ratio:

6 : 2 : 3 : 1 : 3 : 1

late flowering, purple flowers late flowering, white flowers mid flowering, purple flowers mid flowering, white flowers early flowering, purple flowers early flowering, white flowers

OR

2 : 1 : 1 : 3 : 1

late flowering mid flowering early flowering and purple flowers white flowers

Correct offspring genotype with phenotype and ratio;

(b)

- In inheritance caused by multiple alleles, **one gene** is involved in determining the characteristic with **each allele having a discernible/** qualitative **effect** whereas in polygenic inheritance, **several genes** are involved in determining the same characteristic with each allele having an **additive/** quantitative **effect**;;
- In inheritance caused by multiple alleles, the **phenotypes show discontinuous variation/** fall into discrete classes without intermediates whereas in polygenic inheritance, the phenotypes show **continuous variation/** a range of values from 1 extreme to another;;